

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Harvest rate below the maximum sustainable yield; only economically valuable species (mahogany) selectively logged, not clear-felled (1) • Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing; large seed trees and small trees left to grow (1) • Buffer zones with no harvesting or machinery near rivers / wetlands (1) • Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees (1) • Each felled tree tagged and tracked to the retailer for FSC accreditation; rangers control illegal logging; tree nurseries replant over-exploited and ecologically important species (1)	5
01.2	<i>1 mark for each point.</i> • The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it (1) • Employment as rangers, forestry and tourism workers; ecotourists use local restaurants, produce and souvenirs; offcuts given to local carvers (1) • Benefits give the community an incentive to protect the forest; carbon sequestration payments from companies fund management (1)	3
01.3	<i>1 mark for each point.</i> • Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept; timber resource used over a longer period (1) • Disadvantage: more labour-intensive and expensive; more tracks needed per tree (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> - Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity (1) - Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) (1) - Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop (1)	3
02.2	<i>1 mark for each point.</i> - A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) (1) - They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) (1) - Many are in regions where environmental degradation / habitat loss threatens the wild plants (1)	3
02.3	<i>1 mark for each point.</i> - A large population may all descend from a few closely related individuals so they are genetically similar (1) - Similar adaptations mean all are susceptible to the same change (disease, climate) (1) - Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs); so species should be protected across their whole range to retain regional genes (1)	3
02.4	<i>1 mark.</i> Moroma bean (drought resistant, soya-like protein); yeheb tree (drought resistant nuts, poor soils); potato bean (high protein); perennial maize / Kernza (no annual replanting, less ploughing / erosion); salt-tolerant spinach relatives (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take (2) - Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect (2) - Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor (2) - Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected (2)	8
03.2	<i>1 mark for each point.</i> - Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition (1) - Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	
04.1	<i>1 mark for each point.</i> • Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) (1) • Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted (1) • Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion (1)	3
04.2	<i>1 mark for each point.</i> • Control of movement (housing / restricted grazing) so less energy is used in muscle activity (1) • Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature (1) • Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage (1)	3
04.3	<i>1 mark for each point.</i> • Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material (1) • Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) (1) • Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot (1) • Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	<i>Correct answer only.</i> • Its mass (1)	1
05.2	<i>1 mark for each point.</i> • The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced (1) • The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions (1)	2
05.3	<i>1 mark for each point.</i> • The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze (1) • Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away; at much lower pressure there would be no liquid water (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> - Negative feedback responds to any change by re-establishing the previous equilibrium (1) - Only large unusual events overwhelm the feedback so a new equilibrium forms (1) - Example: methane hydrate release at the end of the Permian causing climate change and mass extinction (1)	3
06.2	<i>1 mark for each point.</i> - In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change (1) - Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate (1) - Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather (1) - Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source (1)	4
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Melting permafrost releasing methane (1) - Reduced albedo as ice / snow melt (1) - Methane hydrate release (1) - Forest and peat fires (1) - Faster soil decomposition (1) - Ocean acidification reducing sequestration (1) - Cirrus cloud formation (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
07.1	<i>1 mark for each point.</i> - MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) (1) - LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia (1) - Herbivores are more energy-efficient (lower trophic level) so more food per unit area (1)	3
07.2	<i>1 mark for each point.</i> - Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm (1) - Crossed with normal females (XX) all offspring are XX females (1) - Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently (1)	3
07.3	<i>1 mark for each point.</i> - Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation (1) - Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density (1) - Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification (1) - Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
08.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive (2) • Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light; diversity falls (2) • Hedgerow removal destroys habitat / corridors and nesting sites; loss of natural pest predator habitats increases pest damage; larger fields increase wind erosion; genetic contamination and introduced species also accepted (2)	6
08.2	<i>1 mark for each point.</i> • Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows (1) • These communities depend on the farming continuing (1)	2
08.3	<i>1 mark for each point.</i> • CO₂ from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery; ploughing increases soil respiration (1) • Methane from livestock intestines and rice paddies; nitrous oxide from manure and nitrogen fertilisers (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
09.1	<i>1 mark for each point.</i> • Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud (1) • Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion (1) • Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration (1)	3
09.2	<i>1 mark for each point.</i> • Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) (1) • Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea (1) • Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs (1) • Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds) (1)	4
09.3	<i>1 mark for each point.</i> • Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast (1) • Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) (1) • Replanting and natural regeneration; control of damaging activities; Ramsar sites (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	

Question	Marking guidance	Marks
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
10.2	<i>1 mark.</i> • Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
11.1	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradication, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25